

Continuous PRECICE® AMP Deaminase Assay Kit:
For measurement of 5'-Adenosine monophosphate deaminase (AMPD, AMPDA) activity
For research use only. Not for use in diagnostic procedures

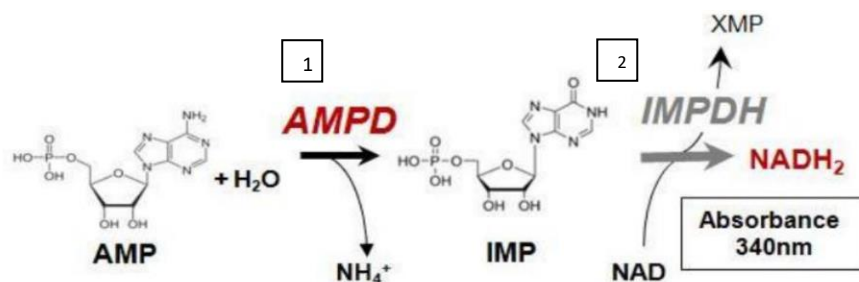
I. Background

AMP deaminase, which catalyses the conversion of AMP to IMP with concomitant release of ammonia, plays a crucial role in regulating adenine nucleotide levels during periods of intense ATP consumption (e.g., exercise). By converting AMP to IMP, AMP deaminase prevents excessive accumulation of AMP and supports maintenance of cellular energy balance. Multiple AMPD isoforms exist, with AMPD1 predominantly expressed in skeletal muscle and AMPD2 in various other tissues. In humans, deficiencies in AMP deaminase—particularly AMPD1 in skeletal muscle—are associated with exercise intolerance.

II. Principle

PRECICE® AMPD Assay Kit is a continuous assay designed to measure AMP-deaminase content in samples. This enzymatic assay is based on a reaction involving Inosine Monophosphate Dehydrogenase (IMPDH).

The principle of the assay is based on the coupling of the following enzymatic reactions:



- (1) In the presence of AMP, AMP Deaminase (AMPD) enzyme catalyzes the formation of IMP;
- (2) In the presence of NAD, IMP is immediately oxidized by a highly active IMPDH in the presence of NAD with simultaneous formation of NADH₂ directly monitored spectrophotometrically at 340 nm.

The assay is developed for measuring AMP deaminase activity in vitro or in cell lysates. *For maximal accuracy*, the assays with cell lysates are run with and without AMP substrate in parallel. The absorbance rate observed in the absence of P-ribose is used as blank and is subtracted from the absorbance rate measured in its presence.

III. Equipments required

- 1) Plate agitator
- 2) Plate reader fitted with a filter 340nm (ex. Labsystems iEMS Reader MF (Thermo), Epoch (BioTec); PerkinElmer).

IMPORTANT:

The following instructions are given to measure the activity of AMP-deaminase enzyme, in a range allowing this measurement by spectrophotometry as described here below. NovoCIB does not guarantee the use of its PRECICE® AMP-deaminase Assay Kit or of one or several of its components, in other conditions than those described in this user manual and/or for other purpose than R&D.

IV. Kit Contents for 24 analysis (2x5mL):

Once dissolved, the reagents should be used on the same day. Two sets of tubes are provided, allowing preparation of either 5 mL or 10 mL of solution.

A standard PRECICE® PRPP-S Assay Kit contains 2 sets of tubes to prepare 2x5mL of reaction mixture.



Each set contains:

- One tube "Cofactor 1";
- One tube "Cofactor 2"
- One tube "Highly active IMPDH", lyophilized;
- One tube "Reaction buffer" (5mL);
- One tube "AMP";
- One tube "IMP";
- One "AMPD-deaminase" (AMPDA, bacterial, positive control)
- one transparent 96-well plate (round-bottom 96-well plate Corning, Costar®, ref. 3797)

IMPORTANT: Use only **autoclaved Milli-Q** water to inactivate ubiquitous phosphatases and to avoid dephosphorylation of P-ribose and PRPP present in reaction mixture

V. Preparation of "Reaction mixture"

1. Thaw "Reaction buffer" at room temperature (do not heat); equilibrate at room temperature;
2. Shortly spin the tubes before opening to recover the powder at the bottom;
3. Add 100µL of phosphatase-free water to IMP powder, agitate to dissolve;
4. Add 100µL of phosphatase-free water to AMP (AMP-deaminase substrate), agitate until complete dissolution of the powder;
5. Add 200µL of phosphatase-free water to lyophilized AMP-DA, vortex vigorously and spin the tube to remove the foam.
6. Quantitatively transfer "Cofactor 1", "Cofactor 2" and "IMPDH" to 5mL of "Reaction buffer".

To do so:

- pipet 1ml of "Reaction buffer" to each of 3 tubes, wait for 5min, mix by inverting or vortex until the powder dissolved;
- transfer by pipetting the content of three tubes back into a vial "Reaction buffer";
- repeat to be sure that all reagent and enzymes of the small tubes and vial are recovered. mix by gently inverting until complete dissolution.

7. Add 50µL of AMP solution to a clean tube followed by addition of 2.5mL of "Reaction buffer"

You have prepared:

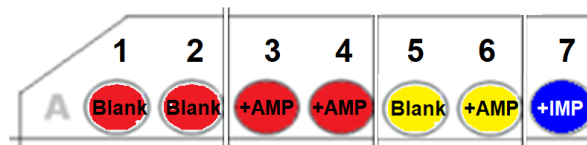
- 2.5ml of "Reaction mixture 1x" without AMP (Blank)**
- 2.5ml of "Reaction mixture 1x with 2mM AMP" (Start)**

VIII. Reaction monitoring for hemolysates, cell lysates or purified/partially purified AMP-deaminase.

Preparation of hemolysates: This protocol was developed with erythrocytes purified from 1mL of peripheral blood by freeze-thawing. Packed erythrocytes (red blood cells, RBCs) were isolated from 1 mL of peripheral blood. The RBC pellet was frozen at -20°C to initiate cell lysis. The frozen pellet was resuspended in 4 mL of ice-cold distilled water (dH₂O), subjected to a second freeze-thaw cycle to enhance lysis. The lysate was clarified by high-speed centrifugation at 13,000 rpm for 5 minutes to remove debris and intact cells.

1. Program plate reader for kinetics absorbance reading every 1min, 37°C;
2. Add 4μL of hemolysate (or cell lysate or PRPP-S from other source) to four wells (example A1, A2, A3 and A4);
3. **Positive control 1 (optional).** Add 4μL of prepared AMP-DA solution per well in two wells (A5-A6);
4. **Positive control 2 (optional).** Add 4μL of IMP solution to empty well A7;
5. Add 200μL of “**Reaction mixture 1x without AMP**” to wells **Blank** (A1, A2, A5) and A7 (IMP);
6. Add 200μL of “**Reaction mixture 1x with 2mM AMP**” to wells “**+AMP**” (A3, A4 and A6) to start the reaction;
7. Agitate and monitor the reaction at 340nm at 37°C for 30min or longer with data collection every minute.

Add 4μL hemolysate per well 4μL AMP-DA



Typical results obtained with RBC lysates are shown on Table 1 / Figure 1.

Table 1.

Time, min	Blank, 4μL RBC lysate	Blank, 4μL RBC lysate	4μL RBC lysate + AMP	4μL RBC lysate + AMP	AMPDA, sans AMP	AMPDA, + AMP	IMP
0	0,781	0,824	0,824	0,852	0,188	0,204	0,578
1	0,783	0,826	0,83	0,858	0,189	0,231	1,295
2	0,783	0,826	0,835	0,862	0,19	0,264	1,909
3	0,784	0,829	0,84	0,868	0,191	0,301	2,427
4	0,785	0,828	0,845	0,873	0,191	0,348	2,827
5	0,787	0,83	0,852	0,88	0,192	0,406	3,103
6	0,787	0,831	0,857	0,886	0,193	0,465	3,268
7	0,789	0,831	0,861	0,89	0,194	0,521	3,334
8	0,789	0,833	0,867	0,895	0,195	0,573	3,395
9	0,791	0,834	0,873	0,901	0,196	0,629	3,422
10	0,79	0,833	0,877	0,905	0,197	0,69	3,446
11	0,792	0,836	0,885	0,912	0,198	0,755	3,446
12	0,792	0,836	0,89	0,917	0,199	0,825	3,446
13	0,793	0,836	0,895	0,923	0,2	0,895	3,446
14	0,794	0,838	0,902	0,929	0,202	0,973	3,446
15	0,794	0,838	0,908	0,935	0,203	1,053	3,446
16	0,793	0,837	0,912	0,939	0,203	1,135	3,446
17	0,795	0,839	0,92	0,946	0,205	1,221	3,446
18	0,796	0,84	0,926	0,953	0,206	1,308	3,446
19	0,796	0,84	0,932	0,959	0,207	1,399	3,446
20	0,796	0,84	0,94	0,967	0,208	1,492	3,446
21	0,797	0,841	0,947	0,974	0,208	1,586	3,446
22	0,798	0,842	0,954	0,981	0,21	1,683	3,446
23	0,798	0,842	0,961	0,988	0,21	1,78	3,446
Absorbance rate per minute	0,0007	0,0008	0,0059	0,0058	0,0010	0,0692	0,5068
Mean absorbance rate per min	0,001		0,006		0,0010	0,0692	0,5068
NADH formation rate, μmols/hour/mL	8,995		71,43				
Protein concentration in well, mg/ml	0,000		0,800				
AMP-DA activity in RBC lysate, nmol/hour/mg			89,29				

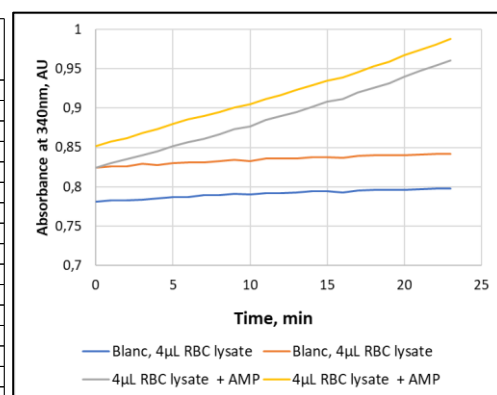


Figure 1. Kinetics of IMP formation from AMP catalysed by AMP-deaminase from hemolysate (4μL of hemolysate per well, 37°C).

X. Calculating the AMP-DA activity in cell lysate (nmols/hour/mg)

1. Calculate the absorbance rate per hour for reaction buffers with AMP (AR_{AMP}) and without (A_{blank}).
2. Calculate Mean AR_{AMP} and Mean A_{blank}
3. Measure the concentration of protein in hemolysates.
4. AMP-DA activity is calculated by the following formula:

$$\text{Activity (nmols/hour/mg of protein)} = (\text{Mean } AR_{AMP} - \text{Mean } AR_{Blank}) * 10^3 / 4.8 / [\text{protein}]$$

Where 4.8 is the absorbance of 1mM NADH at 340nm in round-bottom well of Corning microplate Ref. 3797
[protein], final protein concentration

VII. Calculating the activity of AMP-deaminase (U/ml)

1. Calculate the absorbance rate per hour for reaction buffers with AMP (AR_{AMP}) and without (A_{blank}).
2. Calculate Mean AR_{AMP} and Mean A_{blank}
3. AMPD activity (U/ml) in well is calculated by the following formula:

$$\text{Activity (U/ml)} = (\text{Mean } AR_{AMP} - \text{Mean } AR_{Blank}) * 50 / 4.8$$

Where 50 is a dilution factor (20µL per 1mL) and 4.8 is the absorbance of 1mM NADH at 340nm in round-bottom well of Corning microplate Ref. 3797

IMPORTANT: IMPDH activity (0.150U/ml in well) should be always 2-5 times superior to AMPDA activity. If AMP-DA activity is close or comparable to IMPDH activity, dilute AMP-DA sample and repeat the experiment.

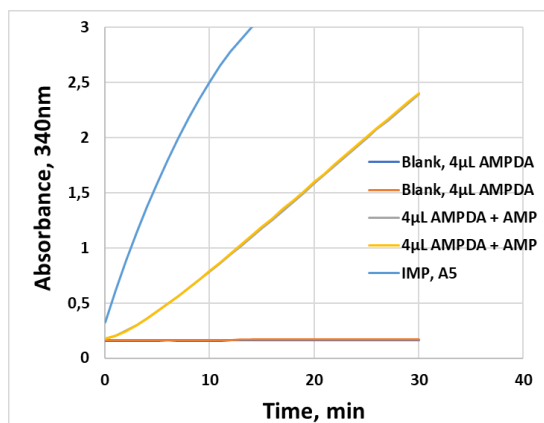


Figure 2. Kinetics of IMP formation from AMP catalysed by AMP-DA provided with the kit (4µL per well).

Table 2.

Time, min	Blank, 4µL AMPDA	Blank, 4µL AMPDA	4µL AMPDA + AMP	4µL AMPDA + AMP	IMP, A5
0	0,162	0,167	0,18	0,176	0,327
1	0,162	0,166	0,206	0,203	0,62
2	0,162	0,166	0,249	0,245	0,889
3	0,162	0,167	0,301	0,298	1,139
4	0,162	0,167	0,36	0,359	1,373
5	0,162	0,167	0,425	0,425	1,594
6	0,163	0,167	0,493	0,494	1,802
7	0,162	0,167	0,563	0,566	1,997
8	0,162	0,167	0,636	0,64	2,181
9	0,162	0,167	0,71	0,715	2,351
10	0,162	0,167	0,785	0,792	2,507
11	0,162	0,167	0,862	0,869	saturated
12	0,163	0,167	0,939	0,948	saturated
13	0,163	0,168	1,017	1,028	saturated
14	0,163	0,168	1,095	1,109	saturated
15	0,163	0,168	1,176	1,19	saturated
16	0,163	0,168	1,256	1,27	saturated
17	0,163	0,168	1,336	1,352	saturated
18	0,163	0,168	1,418	1,432	saturated
19	0,164	0,168	1,499	1,513	saturated
20	0,163	0,169	1,582	1,595	saturated
21	0,163	0,168	1,664	1,676	saturated
22	0,164	0,168	1,747	1,759	saturated
23	0,164	0,169	1,829	1,842	saturated
24	0,164	0,169	1,912	1,924	saturated
25	0,164	0,169	1,993	2,006	saturated
26	0,164	0,169	2,076	2,087	saturated
27	0,164	0,169	2,154	2,168	saturated
28	0,164	0,169	2,235	2,249	saturated
29	0,164	0,169	2,317	2,331	saturated
30	0,164	0,169	2,394	2,406	saturated
Absorbance rate, AU/min	0,000	0,000	0,077	0,077	0,271
Mean absorbance rate, AU/min	0,000		0,077		0,271
AMPDA activity in well, U/ml			0,016		0,055
AMPDA activity in stock solution, U/ml			0,783		